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# Designers' challenges in anticipating and preventing rebound effects: insights from industry interviews

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**ABSTRACT:** Rebound effects occur when sustainability interventions trigger behavioural or systemic responses that offset environmental benefits. This paper explores how designers encounter and seek to prevent them in practice, based on nine interviews with sustainability-oriented practitioners. We identify twelve challenges across micro, meso and macro levels, showing that effective prevention requires aligning behavioural literacy, organisational governance and structural incentives across design contexts.

**KEYWORDS:** rebound effects, design research, behavioural design, systemic design challenges, decision-making

## 1. Introduction

Design for Sustainability has long sought to reduce environmental impacts through improvements in design through efficiency, and more recently, effectiveness (D. C. A. Pigosso et al., 2014; Vilochani et al., 2025). However, the potential of these interventions is frequently undermined by Rebound Effects (RE), which are negative consequences of interventions arising from induced changes in system behaviour (Hertwich, 2005; Pigosso, 2024). Estimates suggest that 50% of the savings from sustainability interventions may be lost due to such effects (Andrew & Pigosso, 2024). While RE are well-theorised within the fields of sustainability science and economics, their implications for design practice remain underexplored, and little is known about how designers themselves encounter, interpret, and attempt to prevent RE in their daily work (Font Vivanco et al., 2022). In the context of this contribution, 'design' refers to the Concept Development phase of the product development process. As defined by Ulrich, (2020), this encompasses a set of activities including the identification of customer needs, the generation of product concepts, and the selection of a final concept for further development.

In the context of Design for Sustainability, the literature on the Circular Economy (CE) and Product-Service Systems (PSS) identifies three persistent gaps:

- RE are most often identified retrospectively, rather than anticipated during the design phase (Das et al., 2023; Müller & Remke, 2025; Zerbino, 2022). There is a gap in the availability of ex-ante tools capable of assessing potential RE at the meso- and macro-levels of CE interventions, Metic & Pigosso (2022).
- Lack of a consolidated and structured overview of existing strategies and tools that enable practitioners to prevent RE during early stages of design (Mckay et al., 2025). While a growing number of studies acknowledge these high-level gaps (Das et al., 2023), few have empirically investigated how these challenges manifest in design practice or what conditions constrain designers' agency to act upon them.
- There are few published studies on the actual prevention of the RE, which is a significant limitation for practical applications (Alfarisi et al., 2023).

As a result, there is a lack of empirical understanding of the practical and organisational barriers designers face when seeking to anticipate and prevent RE. The objective of the paper is thus to



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investigate the following research question: **What are the key challenges from an empirical point of view faced by designers when anticipating and preventing potential RE?**
Through a series of expert interviews across different industry contexts, this study explores how practitioners experience and respond to the behavioural consequences of their design decisions. By providing qualitative insights from the field, this paper contributes empirical grounding to the growing discourse on design for rebound prevention, highlighting the contextual challenges that must be addressed to operationalise preventive approaches in real-world design practice.

## 2. Method

This study employed a qualitative, interview-based approach to explore the real-world challenges faced by designers in anticipating and preventing RE in sustainability-oriented projects. The aim was to generate primary empirical insights into challenges that are insufficiently captured in existing literature. The study was structured in three stages: (1) Selecting and recruiting participants, (2) Designing and conducting the interviews; and (3) Analysing the qualitative data, as following described.

### 2.1. Selecting and recruiting participants

To ensure relevance to RE prevention and diversity across design contexts, participants were identified through purposive sampling. The selection criteria were divided into "need-to-have" requirements and "nice-to-have" criteria attributes (Table 1).

**Table 1.** Criteria and requirements for selecting participants

| Criteria | Requirements |
|---|---|
| Representation across four societal needs (nutrition, mobility, housing, and consumables) | Clear potential for RE within their domain (e.g., projects where behaviour change could offset sustainability gains) |
| Experience with behavioural design or use-phase impacts | Direct involvement in front-end activities of concept development, like design for sustainability, circular innovation, or user-oriented development processes. |
| Positioned early in the design process (concept or framing phases) | Engagement in strategic or decision-making activities related to design |

From an initial pool of 24 identified candidates, nine participants were selected for structured interviews following a two-stage screening process. Candidates were first evaluated against essential 'need-to-have' eligibility criteria, followed by a secondary 'nice-to-have' selection to ensure a diverse range of sustainability-related design roles across industry and consultancy contexts (Table 2).

**Table 2.** Participant overview

| ID | Current Role and Affiliation | Relevant Background and Focus |
|---|---|---|
| P1 | Strategic team lead in city of Copenhagen's department of resource and waste management. | Extensive insight into the practical and strategic challenges of integrating behavioural science within a large municipal setting. PhD focused on climate-friendly consumption in an urban context. |
| P2 | Behavioural Consultant at a behavioural consultancy | Applied anthropologist specialised in behavioural change, direct involvement in testing sustainability interventions, and use of structured behavioural models. |
| P3 | Senior Manager within sustainable and healthy buildings | Expertise in the intersection of sustainable engineering, architecture, and user behaviour within the built environment. |
| P4 | Circular Economy Consultant | Expertise covers eco-design, life cycle assessments, value chain coordination (e.g., Extended Producer Responsibility schemes), and policy work. |

(*Continued*)



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Table 2. Continued

| ID | Current Role and Affiliation | Relevant Background and Focus |
|---|---|---|
| P5 | Partner and co-founder of an anthropological consultancy | Anthropological and strategic perspective on structural change, long-term behaviour, and the dynamics of consultancy work within the green transition. |
| P6 | Behavioural strategist | Expertise in applied behavioural science, and provided a deep, theoretical framework by linking behavioural changes to complex cognitive biases |
| P7 | Senior Sustainability Specialist in the R&D department in MedTech. | Focuses on tech enablement (materials, life cycle assessment (LCA), development opportunities) and supporting new product development while balancing sustainability goals. |
| P8 | Senior Consultant within climate team | Experience as an LCA consultant, with experience on green transition in hospitals |
| P9 | Strategic Designer and Project Manager | Practical experience in creating design support tools intended to guide designers in incorporating behavioural science into the development process |

The participant group offers comprehensive and balanced coverage of all selection criteria and requirements, combining expertise in behavioural science, sustainability-oriented design practice, and strategic decision-making roles across public, private, and consultancy contexts. While representation is strongest in housing, consumables, nutrition, and urban systems (and less direct in mobility), the overall composition ensures comprehensive, multi-level insights into the challenges of anticipating and preventing RE.

## 2.2. Designing and conducting the interviews

The interview guide was designed to elicit both retrospective reflections on past design experiences and forward-looking perspectives on preventive tools for RE. The interview guide comprised questions related to uncovering the context, barriers, and responsibility of designers, and was developed iteratively to ensure that the questions were comprehensible, non-leading, and allowed for both practical and reflective responses. The structure struck a balance between exploratory depth and comparability across participants. The protocol was pilot-tested and remained consistent throughout all 9 interviews to ensure comparability. The total duration of the data collection phase spanned approximately six weeks. Each interview lasted 1.25 hours and was attended by two researchers.

## 2.3. Analysing the qualitative data

Data analysis followed an iterative, inductive coding process, and, composed of three main steps: (1) Initial familiarisation with the data through reading and summarisation; (2) Identification of key themes and recurring barriers related to designers' ability to anticipate or prevent RE through colouring of similar challenges; (3) Clustering of challenges into higher-level categories (e.g., organisational, methodological, knowledge-related). The coding and synthesis were performed manually by a single researcher to maintain close interpretive engagement with the data. The resulting set of challenge areas were later reviewed by an expert, and forms the empirical basis for the findings presented in Section 3.

# 3. Results

Throughout the data collection, 12 unique challenges faced by designers when anticipating and/or preventing RE were consolidated. Each challenge is accompanied by an overall description, followed by a representative quote from the interviews.

**C1. Difficulty in anticipating the complexity of human behaviour**

Across all nine interviews, participants emphasised that human behaviour remains one of the most unpredictable and difficult factors to anticipate in design for sustainability. Designers have limited behavioural awareness of the complexity of human behaviour (e.g., understanding and preventing cognitive biases). As a result, designers frequently underestimate the behavioural complexity underlying RE, and do not assume these "illogical behaviours". As P2 states: "*RE are difficult precisely because*



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*they're not logical. When you are deep into developing behavioural solutions, it seems quite distant that someone would do something worse because they had done something good"* (moral licensing) (Van der Loo & Pigosso, 2024). This finding highlights the challenges of limited behavioural awareness and the irrational nature of human decision-making in identifying and addressing RE during the design process.

**C2. Missing education in users on the hidden environmental impacts**

When environmental impacts are invisible or abstract, users struggle to connect their everyday actions with their ecological consequences. Without tangible cues or feedback, it becomes harder for users and designers to recognise how well-intentioned solutions may trigger RE, allowing environmentally harmful behaviours to persist unnoticed. "*They don't realise how much CO2 emissions and how much environmental impact a building has. . . . But you just don't see it as a user - some of the impacts I think are hidden . . . And if you don't have enough knowledge, if you don't get the transparency of all those impacts, then you just don't see them. People just don't realise until they are actually told and given some examples. So I think it is about education. It is about sharing more and transparency" (P4).*

**C3. Difficulty of measuring behavioural shifts**

Obtaining honest data is difficult because people tend to act according to instructions when observed. Behavioural effects often remain hidden until real use is observed. P8 mentions: *"It is extremely challenging for project teams to gather accurate data on real-world usage and problematic habits. When people know they are being observed, they tend to perform according to prescribed procedures (e.g., following hygiene protocols rigorously)"*. In a hospital project, reported procedures differed from nurses' actual behaviour. If the change had been designed solely from the office, relying on reported practices, the intervention would have appeared successful. Yet, field observation showed that nurses acted differently than they claimed, leading to this inadvertent outcome. Thus, *"Rebound-like effects emerge only when designers dig a few spades deeper" (P8).* Finally, capturing behavioural shifts is also prone to temporal distortions such as regression to the mean or novelty fade. Regression to the mean occurs when significant behaviours or performance levels naturally move closer to the average over time, making initial improvements or declines appear less stable than they are (Yu & Chen, 2014). Novelty fade describes the tendency for initial enthusiasm following a new intervention to diminish as the novelty wears off, leading to a gradual return to habitual behaviour (Rodrigues et al., 2022). In a food waste study, (P2) noted that individuals who were already 'high performers' before the intervention showed a fall in sorting performance afterwards.

**C4. Narrow evaluation scopes, time frames, and insufficient tools to capture behavioural shifts**

Design success is often measured within narrow scopes, such as short timeframes, that overlook indirect or long-term effects. Current tools and processes (e.g., LCA, ecodesign) rarely account for behavioural spillovers or systemic feedback, leaving many RE invisible in evaluation and decision-making. Currently, there is a lack of clear procedures for identifying potential behavioural shifts early in design phases, and the funding is not there either - *"We only measure what the brief pays for."* (P2). While LCA are used to model effects, current industry practice heavily favours attributional LCA, which is fast and simple but sets tight system boundaries, and systemic perspectives, which dictate RE, are often omitted from tools because they are considered too complicated. As (P8) states: *"These cases are done relatively quickly and on a uniform basis, and that's why you use an attributional approach. And that means contractors or consultants learn it in this way . . . They use LCA in a way that they multiply two things together in a spreadsheet, and that's it, where they don't have this system perspective"*.

**C5. Behavioural design is seen as a mitigation measure**

This challenge revolves around how organisations deal with behavioural "fixes". In many cases, behavioural issues are addressed after implementation, ex-post rather than ex-ante. P6 contrasts "behavioural design" with "behavioural strategy" by stating that the former often becomes a reactive fix for poor design: *"Behavioural design is typically to mitigate someone who has made a piece of crap design. So we have to take action to prevent it from causing so much damage. A behavioural strategy, that is, you start by getting a deep understanding of users' lifestyles and contexts, and what do our products mean in this life form from the beginning? This is what Silicon Valley and those types of places are super good at understanding, and they're 100 times further ahead of us"*. The approach for preventing RE must then take a proactive, rather than a reactive approach, to make real progress.

**C6. Low organisational maturity and designers' agency**

Designers often lack the mandate and decision-making power to influence strategic priorities. In most organisations, top management agendas focused on growth and short-term performance dominate,



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leaving little room for addressing complex or indirect issues, such as RE. As a result, many companies, still in the early stages of sustainability maturity, perceive rebound prevention as too abstract or premature to integrate into their current processes. *"Strategic decisions are typically made by those discussing long-term profitability and money (top management), not designers"* (P3). Preventing RE is thus a matter of organisational maturity, not only on a behavioural-, but also on a sustainability scale - *"There are some baby steps you need to take with the companies before you really get up to running speed (on sustainability). And here, when you come in and focus on, let's say, behavioural aspects, I think it sounds quite sophisticated to me. There are some that even don't know why they should do a life cycle assessment"* (P7).

**C7. Fragmented and linear design processes**

The traditional linear design process (e.g. developer to architect to engineer) inherently fragments perspectives, leading to inefficient solutions. As (P3) notes: *"you don't see all the different experts at the same table from the beginning"*. This fragmentation often results in a series of late-stage adjustments or "small fixes," which ultimately produce outcomes that are *"less usable or less clear for users."* Such sequential processes reinforce siloed thinking, preventing professionals from recognising how one intervention may influence another. *"Behavioural considerations typically come too late, after product development, when design levers are already fixed. When they are included late in the design process, this leads to a mismatch between design intent and actual usage"* (P3). In contrast, a *'transformative partnership'* may help prevent RE by enabling a broader range of perspectives to anticipate potential issues early in the process. As (P1) emphasised, the responsibility for fundamental change cannot lie with a single person or role, and we must *"understand nuances"* and recognise the unique contributions of individuals like *"an anthropologist, an economist and a political scientist"*.

**C8. Limited priority given to behavioural outcomes in design projects**

Many designers and organisations only focus on product aspects such as technical efficiency rather than behavioural outcomes. As a result, they are largely unaware of RE or how behaviour arises through real world use. This highlights a need for targeted training in identifying "blind spots" - their unknown unknowns that lie outside their current awareness and expertise. *"I try to help all the technical people understand that their projects influence behaviour, because historically our field (of waste management) hasn't really considered that. But now we are quite far ahead of understanding this irrational behaviour. There are definitely still technical solutions being developed that people just assume will work on their own, regardless of how they're used"* (P1). Companies that do not adopt a user-centric approach also lack these capabilities on a fundamental level, as P9 mentions: *"Many companies struggle to genuinely adopt the consumer's perspective. When they attempt to do so, they often project their own assumptions onto users. There is a huge importance of making designers and organisations aware that people think and act differently from themselves, and that individuals' varying willingness to act strongly influences whether sustainable behaviours will actually occur"*.

**C9. Conflicting expectations of the behavioural change project**

Clients often lack understanding of what behavioural design can realistically achieve, leading them to resist investing in the in-depth ethnographic research needed to uncover real user motivations and contextual behaviours. As a result, projects rely on limited data and rarely capture how people actually behave after launch. Several participants explained that clients tend to prefer quick, surface-level methods, such as surveys or simple nudges, because they appear faster and cheaper. *"Behavioural design consultancies often lose projects to marketing agencies that promise large-scale change through catchy slogans and visual branding"* (P2). Their consultancy, by contrast, only promises to change 'a small behaviour', yet this evidence-based modesty can appear less attractive to clients expecting rapid or total transformation. This mismatch between client expectations and the realistic scope of behavioural interventions hampers the integration of nuanced, rebound-aware design practices. This issue is compounded by financial and contractual constrains, as (P5) illustrated, *"I work by the hour, so that means, should we add an extra phase to eliminate rebound? . . . Does the customer have the money?"*. Her reflection highlights how resource limitations prevent sustained engagement, which is necessary to assess long-term outcomes. Expanding on this temporal tension, P5 emphasized that meaningful sustainability transitions unfold over much longer timescales. *"I look almost at entire generations when I think about how the green transition will happen. I don't think about the next six months . . . . So for me, it's also really about daring to look at it with the long-term perspective and say, it's the long, hard pulls"*. Her reflection illustrates a key tension between the short-term orientation of design projects and the inherently long-term nature of sustainability transitions.



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**C10. RE is seen as an inconvenient truth for designers**

RE are an "inconvenient truth", which makes them easier to ignore, also for the behavioural designers. They have already managed to fix certain behaviours, and RE can thus undermine the perceived success of sustainability initiatives. As (P2) mentions: *"if you were to look at it critically, there is also some kind of moral licensing in saying, when we work to reduce a CO2 footprint, then that must also be good enough. It can't also be our job that people aren't idiots when they have minimised their CO2. And then, in reality, it's an inconvenient truth - when you try to help, and then if it really just fuels something even worse. So that's not something you seek out in reality either"*.

**C11. Structural and policy constraints limit sustainable progress**

Legal and sectoral frameworks often limit designers' ability to address RE. Municipalities are mandated to manage waste rather than reduce it, while safety and sterility requirements in sectors like healthcare override sustainability concerns. At the same time, weak regulation of marketing fuels continued overconsumption. As P5 states, *"The state/EU has a huge responsibility to regulate marketing that pushes 'consumerism' that is completely out of control. This could be CO2 taxes that are 'significant' rather than symbolic, and outright bans on certain types of consumption. Without this high-level intervention, RE are extremely difficult to overcome because people will always find a loophole or justification for unsustainable behaviour"*. Her statement shows that while design can contribute to behavioural change, enduring progress ultimately depends on supportive institutional and regulatory frameworks. Building on this, P6 notes that: *"Some rebounds are too systemically reinforced to nudge away; pricing and standards are sometimes the only stable fix. Designers should know when to escalate to policy levers."*. Certain sustainability challenges extend beyond the reach of design interventions alone, requiring coordinated policy measures and structural reform to prevent RE at scale. *"I think we can't sit around and wait for companies to self-regulate in this regard"* (P4).

**C12. Cultural normalisation of increased comfort and consumption**

Designers operate within cultural and economic systems that equate comfort, novelty, and material consumption with progress and success. Consequently, increased consumption is still widely regarded as a desirable design outcome, even when it provides no additional functional or social benefit. As (P6) remarked, *"Why should we live less comfortably than we could, and why should we spend less than we can afford?"* This sentiment illustrates how cultural expectations of continuous improvement and comfort constrain the legitimacy of "less" as a design ambition. Participants further reflected on the psychological roots of these tendencies. As (P9) observed, *"we are evolutionarily programmed to seek comfort and shortcuts. Our brains are always looking for the easiest way."* This inherent bias towards effort reduction and pleasure-seeking reinforces societal patterns of overconsumption, making it difficult for designers to promote moderation or sufficiency as attractive and aspirational goals. Together, these reflections highlight that the rebound challenge is not only structural but also deeply cultural, embedded in collective notions of progress, comfort, and the good life.

## 4. Discussion

These 12 challenges illustrate why preventing the Rebound Effect (RE) remains so difficult in sustainable design. While comparing these findings to broader theory remains a task for future research, this study intentionally focuses on filling the gap in empirical evidence. This focus does limit generalisability, specifically regarding sampling and analyst bias, but it provides a necessary, grounded perspective. Together, these barriers reveal a multi-layered landscape spanning individual cognition, organisational priorities, and societal structures. To understand the implications of this landscape, the findings can be interpreted across three levels of intervention:

- **Micro level:** focuses on the cognitive and perceptual challenges faced by individual designers and users. These are psychological and informational challenges.
- **Meso level:** encompasses organisational systems, design processes, and evaluation practices which are directly within firms' or designers' sphere of influence.
- **Macro level:** addresses systemic and cultural conditions beyond individual or organisational control, yet critical for enabling lasting change.

The clusters of challenges within each level are presented in Table 3.



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**Table 3.** Challenge overview, %XX = Percentage of participants experiencing the challenge

| Level | Cluster | Description | Challenges |
|---|---|---|---|
| **Micro** | *Human and behavioural foundations* | Designers and users struggle to anticipate irrational human behaviour and to recognise hidden environmental consequences. | C1(66%), C2(100%) |
| **Meso** | *Methods and evaluation frameworks* | Measuring behavioural shifts and RE is difficult due to poor tools, narrow timeframes, and missing feedback loops. | C3(44%), C4(100%) |
| | *Organisational maturity and behavioural integration* | Organisations focus on efficiency and fixes, while designers lack agency and structures for proactive rebound prevention. | C5(55%), C6(55%), C8(55%) |
| | *Design processes and collaboration* | Siloed, sequential workflows limit early cross-disciplinary integration of behavioural perspectives. | C7(55%) |
| | *Conflicting priorities and denial* | Short-term goals, client pressures, and denial of RE prevent long-term behavioural learning. | C9(55%), C10(55%) |
| **Macro** | *Structural and policy constraints* | Sectoral regulations and weak policy incentives restrict designers' ability to address RE at scale. | C11(33%) |
| | *Cultural developments* | Societal norms equating progress with comfort and novelty perpetuate rebound-prone design goals. | C12(55%) |

At the micro level, the challenges reveal the psychological and perceptual boundaries of both designers and users (C1). Human behaviour is inherently complex and often irrational, shaped by cognitive biases such as moral licensing. Designers outside of behavioural firms rarely have the tools or frameworks to anticipate such dynamics (C2). These findings underline the need for greater behavioural literacy in design education and practice. Yet, as later levels show, such awareness alone is insufficient if organisational structures and incentives do not enable its integration.

At the meso-level, RE often remain invisible by design. Organisations do not have the correct methods and evaluation frameworks to measure behavioural shift, such as RE (C3, C8). Early-phase tools and success metrics often fail to capture behavioural shifts, while the absence of post-implementation monitoring hinders learning from real-world use (C8). Current design processes and evaluation practices continue to favour efficiency, aesthetics, and short-term deliverables over systemic and behavioural outcomes – issues that are clearly explained and demonstrated in recent studies (Andrew et al., 2024; Guzzo et al., 2024; Van der Loo & Pigosso, 2024). Challenges such as limited client understanding (C9), short-term contracts (C10), and fragmented linear design processes (C7) illustrate how organisational conditions restrict the capacity for long-term, cross-disciplinary learning.

This is also a symptom of the lack of organisational maturity in terms of behavioral integration (C4, C5, C6). Behavioural design remains an add-on, applied reactively to fix problems rather than proactively to shape systems (C5). Financial and contractual structures further reinforce this short-termism (C9). Behavioural consultancies face pressure to deliver quick, measurable outcomes rather than nuanced, long-term insights (C10), and those who succeed with delivering effective behaviour change would rather look at what they've achieved, than to address the inconvenient truth that is RE (C10).

While design is often positioned as a key enabler of sustainable transitions, its influence is constrained by organisational hierarchies and economic priorities, and several participants questioned the extent to which designers can realistically drive systemic change toward sufficiency. As (P4) reflected: *"It's very hard to say, designers should be the ones driving this, because they tend not to have that type of influence. The people who matter are the ones discussing the money . . . If you can convince them that it is in their best interest to get people to consume less . . . that's where you can have an impact. And then suddenly the designing becomes easy at that point, I think!"*.

Moreover, the maturity levels of sustainable corporate entrepreneurship highlight that firms vary significantly in sustainability maturity, and those lower on the maturity scale tend to lack strategic integration of sustainability and cross-functional influence (Schönwälder & Weber, 2023). Designers and sustainability leads often lack the mandate to influence strategic priorities, while top management continues to prioritise growth and profitability over sufficiency (C6). This challenge positions rebound prevention not merely as a design challenge but as an issue of organisational governance and strategic



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alignment (C6). Preventing RE requires reframing the design brief to include higher-order, systemic effects and securing decision-maker buy-in for longer time horizons (C9).

The findings also point toward promising avenues, such as transformative partnerships, where multiple experts are "at the same table from the beginning" enable shared understanding of behavioural and systemic risks (C7). Such "transformative partnerships" can break the silos that sustain RE and distribute responsibility across professional boundaries.

At the macro level, participants emphasised that many rebound dynamics are embedded in broader structural and cultural conditions that extend beyond the reach of individual designers or firms (C11, C12). Legal mandates, sectoral standards, and weak regulation of marketing collectively sustain consumption-driven systems (C11). Design interventions alone cannot counteract systemic incentives for overconsumption. Structural reforms, such as stronger sufficiency-oriented policy frameworks, pricing mechanisms, and marketing restrictions, are needed to create an enabling context where sustainable design strategies can have a lasting impact. Beyond structural constraints, the interviews highlight the deep cultural normalisation of comfort, novelty, and growth (C12). Participants described how the dominant narrative of progress as "more" rather than "enough" undermines sufficiency-oriented ambitions. It exposes the cultural underpinnings of RE, suggesting that preventing them requires not only new design methods or policies, but a redefinition of what constitutes a good and desirable life.

Taken together, the challenges reveal that rebound prevention cannot be solved at a single point of leverage (de Oliveira et al., 2024). While behavioural literacy and new design tools are essential, they must be supported by education, organisational maturity, and financial structures that allow for long-term evaluation, and macro-level frameworks that realign cultural and economic incentives. The findings echo Meadows' (1999) notion that higher-order leverage points, such as shifting mindsets, goals, and paradigms, offer the most durable pathways for change. In practical terms, this means reframing design for sustainability from delivering efficient products to cultivating systems that support sufficiency, transparency, and long-term learning. It also demands cross-level collaboration: designers, managers, and policymakers must jointly define what "success" looks like beyond efficiency metrics. Only by addressing the three challenge levels together can design begin to prevent RE rather than merely respond to them. As Lowe et al. (2025) state: *"In this light, circular economy rebound is not merely a technical obstacle: it is a mirror held up to the values, priorities, and structures of our economic system, and an invitation to imagine alternatives"*.

## 5. Conclusion

This study contributes empirical and theoretical advances to the emerging field of design for rebound effect (RE) prevention. Empirically, it surfaces twelve interconnected challenges that designers face across micro (human and behavioural), meso (organisational and institutional), and macro (structural and cultural) levels. Theoretically, the paper demonstrates that effective prevention relies on aligning behavioural strategies with organisational governance and policy contexts, i.e., transitioning from isolated "behavioural design" fixes toward a system-level behavioural strategy.

The study is based on a small, purposively sampled set of nine interviews and a qualitative, context-rich analysis. While this affords depth, it constrains statistical generalisability and may under-represent sectors or maturity levels not included in the sample. The findings should therefore be read as analytic generalisations that can be carefully transferred to comparable contexts. Building on these insights, future work should prioritise the following: by understanding the depth of these challenges, they can be translated into rebound prevention support, which can then be tested in real company contexts for evaluating usability, integration into decision-making processes, and the effects on design choices and post-launch behaviour. This requires broad validation across various sectors (e.g., mobility, nutrition, housing, and consumables) to assess robustness and boundary conditions.

Overall, preventing RE requires synchronised action across levels: by enhancing behavioural literacy for designers and feedback visibility (micro), by reshaping briefs, metrics, and contracting for long-term learning (meso), and by aligning incentives through structural and cultural change (macro). Only by coordinating these interventions can we move from reactive fixes to proactive, system-level prevention.

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